Solution

By definition, we have $|x+3| = \begin{cases} x+3 & \text{if } x \geq -3 \\ -x-3 & \text{if } x < -3 \end{cases}$

See Fig. 9.7 for the graph of $y = |x+3|$ in $-4 \leq x \leq 4$.

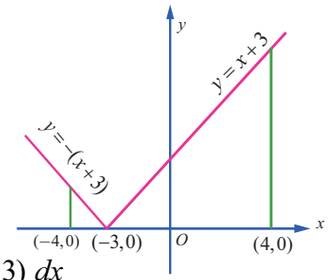


Fig. 9.7

$$\therefore \int_{-4}^{4} |x+3|\, dx = \int_{-4}^{-3} |x+3|\, dx + \int_{-3}^{4} |x+3|\, dx = \int_{-4}^{-3} (-x-3)\, dx + \int_{-3}^{4} (x+3)\, dx$$

$$= \left[-\frac{x^2}{2} - 3x\right]_{-4}^{-3} + \left[\frac{x^2}{2} + 3x\right]_{-3}^{4}$$

$$= \left(-\frac{9}{2}+9\right) - \left(-\frac{16}{2}+12\right) + \left(\frac{16}{2}+12\right) - \left(\frac{9}{2}-9\right) = \left(\frac{9}{2}\right) - 4 + 20 + \left(\frac{9}{2}\right) = 25.$$

Next, we give examples to illustrate the application of Property 5.

Example 9.16

Show that $\int_0^{\frac{\pi}{2}} \frac{dx}{4+5\sin x} = \frac{1}{3}\log_e 2$.

Solution

Put $u = \tan\frac{x}{2}$. Then, $\sin x = \frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}} = \frac{2u}{1+u^2}, du = \frac{1}{2}\sec^2\frac{x}{2}\,dx \Rightarrow dx = \frac{2du}{1+u^2}$.

When $x = 0, u = \tan 0 = 0$. When $x = \frac{\pi}{2}, u = \tan\frac{\pi}{4} = 1$.

$$\therefore I = \int_0^{\frac{\pi}{2}} \frac{dx}{4+5\sin x} = \int_0^1 \frac{\frac{2du}{1+u^2}}{4+5\left(\frac{2u}{1+u^2}\right)} = \int_0^1 \frac{du}{2u^2+5u+2} = \frac{1}{2}\int_0^1 \frac{du}{u^2+\frac{5}{2}u+1}$$

$$= \frac{1}{2}\int_0^1 \frac{du}{\left(u+\frac{5}{4}\right)^2 - \left(\frac{3}{4}\right)^2} = \left[\frac{1}{2}\times\frac{1}{2\times\left(\frac{3}{4}\right)}\log\left(\frac{\left(u+\frac{5}{4}\right)-\frac{3}{4}}{\left(u+\frac{5}{4}\right)+\frac{3}{4}}\right)\right]_0^1 = \frac{1}{3}\left[\log\left(\frac{u+\frac{1}{2}}{u+2}\right)\right]_0^1 = \frac{1}{3}\log 2.$$

Note

To evaluate anti-derivatives of the type $\int \frac{dx}{a\cos x + b\sin x + c}$, we use the substitution method by putting $u = \tan\frac{x}{2}$ so that $\cos x = \frac{1-u^2}{1+u^2}, \sin x = \frac{2u}{1+u^2}, dx = \frac{2du}{1+u^2}$.

Example 9.17

Prove that $\int_0^{\frac{\pi}{4}} \frac{\sin 2x\, dx}{\sin^4 x + \cos^4 x} = \frac{\pi}{4}$.

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